MILITARY SPECIFICATION

CONTROL INDICATOR C-8157( )/ARC

1. SCOPE

1.1 The specification covers one type of control unit which provides control over the operational modes of a speech security device and is designated Control Indicator C-8157( )/ARC. (See 6.1 and 6.4)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566	Box, Folding, Paperboard
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-B-676	Box, Setup
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-S-760	Strapping, Nonmetallic, and (Connectors)
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-B-43291	Box, Fiberboard, Corrugated, Double-Wall, Weather Resistant

FSC 5821

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 in. x 48 in. Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-704	Electrical Power, Aircraft, Characteristics and Utilization of
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

DL-SM-A-535614	Control-Indicator C-8157( )/ARC
SC-A-678003	Burn-In Process Procedure for

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Description. - Control Indicator C-8157( )/ARC is a separately housed unit containing one (1) circuit breaker and three (3) switches, three (3) indicator lamps, and two (2) panel lamps. The switches and lamps are connected to the receptacle on the rear of the control indicator which provides for plain (non-cipher), cipher, retransmission and a remote cipher lamp. The control indicator is flush mounted by means of Dzus fasteners located on the sides of the panel mounting plate. The unit is capable of operating from a power source of 27.5 $\pm$ 0.5 volts dc which is normally obtained from the main power of aircraft.

3.2 Construction. - The units on order shall be constructed in accordance with the following drawing and data list:

DL-SM-A-535614 Control Indicator C-8157( )/ARC

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3.3 Preproduction samples. - The contractor shall furnish four (4) preproduction samples of units on order for approval, if required by the invitation for bids and contract. (See 4.3 and 6.2)

3.4 Cleaning. -

3.4.1 Parts. - After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.4.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.5. Controls. - All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition.

3.6 Finish, protective (See 4.4). - The units shall be given protective finish in accordance with Specification MIL-F-14072 and the equipment drawings. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch-up after mounting. The final paint film on Type 1 surfaces shall be Color No. 37038 (lusterless black) conforming to Specification MIL-F-14072.

3.7 Securing of parts. -

3.7.1 General. - Brackets, lugs, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.7.2 Securing of parts by threaded fasteners. - Nylok or other screws and nuts with plastic devices and Loctite or similar sealants shall not be subjected to temperatures in excess of 250°F (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

3.8 Marking (See 4.4). -

3.8.1 General. - Marking shall conform to Specification MIL-M-13231. Front panel marking shall be Group 1 as described in that specification.

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3.8.2 Visibility. - Whenever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the unit(s).

3.8.3 Serial numbers. - The following unit(s) shall be serial numbered:

Control Indicator C-8157( )/ARC

3.9 Soldering. -

3.9.1 Solder. - Solder shall conform to the equipment drawings.

3.9.2 Acid or acid salts. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning or electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.9.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystalized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the unit(s); flux being spread over a larger area, or damage to the unit. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.10 Wiring and cabling. - Wiring and cabling shall be neat and sturdy.

3.10.1 Cabling. - Insulated wires shall be formed into cables except where operation of the unit(s) would be adversely affected thereby or where it is physically impracticable, as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance.

3.10.2 Slack. - Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

a. To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

b. To enable parts to be removed and replaced during servicing without disconnecting other parts.

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c. To provide for at least two replacements of the part to which the wire or cable is connected, except where uninsulated wires are used.

3.10.3 Protection. - Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges.

3.10.4 Support. - Wire and cable shall be properly supported and secured to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (1) during and after subjection of the unit(s) to specified service conditions or (2) after service or repair of the unit(s) in a normal manner.

3.10.5 Clearance. - Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.10.6 Splicing. - Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.10.7 Connections, general. - Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.11 Tropicalization of materials. - Materials shall be treated as follows:

3.11.1 Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.11.2 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transporation, operation, or maintenance of the unit(s), or during use of the finished items when used for the purpose intended.

3.11.3 Flexibility. - Treatment shall not affect the flexibility of treated materials, to the extent that the unit(s) may fail to meet specified requirements when subjected to specified service conditions.

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3.11.4 Statement of treatment. - The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 3.12.7 and 6.2)

3.12 Service conditions (See 4.7). - The unit(s) shall meet the service conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as otherwise specified herein.

3.12.1 Temperature (See 4.7.1). -

3.12.1.1 Operating. - Operable conforming to limits specified from -65°F to +160°F for continuous operation. The unit(s) shall be considered Class 2, except when otherwise specified.

3.12.1.2 Non-operating. - Exposure to the temperature of -80°F and +203°F.

3.12.2 Altitude (See 4.7.2). - Operational during the required temperature altitude steps ranging from -65°F at 70,000 feet through +95°F at 70,000 feet.

3.12.3 Relative humidity (See 4.7.3). - Up to 100 percent relative humidity including conditions wherein condensation takes place in the form of water. This test shall be performed over a 10 day cycle.

3.12.4 Sand and dust (See 4.7.4). - The unit(s) shall withstand exposure to sand and dust particles as encountered in desert areas.

3.12.5 Vibration (See 4.7.5). - The unit(s) shall meet the vibration requirements of Specification MIL-E-5400 (Curve IV) for combination aircraft and helicopter equipment.

3.12.6 Shock (See 4.7.6). -

3.12.6.1 Benchhandling (See 4.7.6.1). - The unit(s) shall be capable of full specification performance after subjection to the required numbers of drops.

3.12.6.2 Equipment (See 4.7.6.2). - The unit(s) shall meet the equipment shock test. The unit(s) shall be capable of meeting full specification performance after the shock test.

3.12.7 Fungus (See 4.7.7). - The unit(s) shall show no evidence of viable fungus or corrosion after being subjected to 28 days of fungus test.

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3.12.8 Salt fog (See 4.7.8). - The unit(s) shall be exposed, in normal operating configuration, to salt fog for 48 hours and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish.

3.12.9 Explosive conditions (See 4.7.9). - The unit(s) shall meet the explosion proof requirement per MIL-E-5400.

3.13 Electrical-requirements (See 4.8). -

3.13.1 Input power (See 4.8.1). - The unit(s) shall operate directly from a nominal 27.5 dc aircraft power system and shall meet the requirements of paragraphs 5.2 and 3.17 of Standard MIL-STD-704. With reference to 5.2.3.2 of MIL-STD-704, cutoff types of transient suppressors shall not be used.

3.13.2 Connector resistance (See 4.8.2). - The connector (J-1) resistance shall be as indicated when measured in the sequence as shown:

Pins	Resistance Reading in ohms
D to case gnd & each lamp shell	0
D to A (CB-1: off)	infinity
D to B (SW-1: plain)	30 ohms to 100 ohms
D to C (SW-3: off)	infinity
D to E	0
D to F	0
D to G (SW-1: plain)	60 ohms to 200 ohms
D to H	30 ohms to 100 ohms
D to J (SW-1: plain)	30 ohms to 100 ohms
D to J (SW-1: cipher)	0 - 0.1
D to K (SW-2: REG)	infinity
D to K (SW-2: RE-X)	0 - 0.1
A to B (CB-1: off)	infinity

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Pins	Resistance Reading in ohms
A to B (CB-1: on)	0.2 - 0.3 (internal resistance CB-1)
B to C (SW-3: off)	infinity
B to C (SW-3: on)	0 - 0.1
B to G (SW-1: plain)	90 ohms to 300 ohms
B to G (SW-1: cipher)	0 - 0.1
B to J (SW-1: cipher)	30 ohms to 100 ohms
B to J (SW-1: plain)	0 - 0.1

3.13.3 Lamp operation (See 4.8.3). - With the switches in the indicated positions and with 27.5 volts dc applied, the panel illumination lamps, indicator lamps and test lamps shall light to normal brilliance. The covers of the indicator lamps when rotated clockwise shall decrease the brightness.

Switch	Position
Plain-Cipher (SW-1)	Plain
Power-On (CB-1)	On
REG-RE-X (SW-2)	RE-X
Zeroize (SW-3)	On

3.13.4 Circuit breaker, voltage drop (See 4.8.4). - With the switches in the positions indicated in 3.13.3 the voltage drop across the "Power-On" switch shall not be greater than 0.5 volts dc.

3.13.5 Switch voltage drop (See 4.8.5). - With the switches in the position indicated in 3.13.3, the voltage drop from Pin E to Pin C and from Pin E to J shall not be 0.1 volts dc less than the voltage from Pin E to Pin B.

3.13.6 Circuit breaker, trip-out (See 4.8.6). - The switch shall perform its required function after an applied voltage trips the circuit breaker to over-load and the switch has been placed back to "On" position.

3.14 Interchangeability (See 4.9). - Like units, assemblies, and replaceable parts shall conform to Requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable, without modification of such items or of the unit(s).

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3.15 Burn-in, debugging (See 4.6). - Prior to acceptance tests herein, each unit on order shall be capable of meeting the requirements herein after subjection to the debugging of 4.6.

3.16 Operational inspection (See 4.10). - Unit(s) shall be subjected to the operation inspection as specified, after applicable processes, inspection and tests are completed, to determine that each unit is operable prior to packaging.

3.17 Technical literature and spare parts. - Technical literature and spare parts shall be furnished as specified in the contract. Spare parts shall be identical to corresponding parts in the unit(s) furnished on order. (See 6.2)

3.18 Workmanship. - The unit(s) shall be manufactured and assembled in accordance with the equipment drawings and applicable portions of the following requirements and paragraphs herein:

Soldering	Requirement 5 of MIL-STD-454
Workmanship	Requirement 9 of MIL-STD-454
Fastener Hardware	Requirement 12 of MIL-STD-454
Cleaning	3.4
Controls	3.5
Securing of Parts	3.7
Soldering	3.9
Wiring and Cabling	3.10
Tropicalization	3.11

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract a purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

a. Preproduction inspection (does not include preparation for delivery). (See 4.3)

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b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection:

(1) Quality conformance inspection of unit(s) before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.12)

4.3 Preproduction inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B and Group C (see Tables I and II and III respectively). The preproduction inspection will normally be performed in this order: (1) vibration, (2) shock, benchhandling and (3) shock, equipment, other preproduction inspection may precede, follow, or be interspersed between the foregoing.

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery:

Item	Where Required
Finish	3.6
Marking	3.8

4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. - This inspection, including sampling, shall conform to Table I and the inspection procedures of Standard MIL-STD-105 using the general inspection levels. Group A inspection shall be performed in any order which is satisfactory to the Government except that the operational inspection (4.10) shall be last. (See 6.3)

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Table I - Group A Inspection

Inspection	Req. Para.	Insp Para.	AQL Major	AQL Minor
Visual and Mechanical:	3.18	4.11		
C-8157( )/ARC (less cover)			1.0%	4.0%
C-8157( )/ARC (assembled)			1.0%	4.0%
Electrical Tests:				
Lamp operation	3.13.3	4.8.3	*1.0% for combined group	
Circuit breaker, voltage drop	3.13.4	4.8.4		
Switch, voltage drop	3.13.5	4.8.5		
Operational inspection	3.16	4.10	0.01%	

*All electrical defects are considered major.

4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table II and to the procedures for small sample inspection of MIL-STD-105 using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.3 Order of inspection within group B. - Group B inspection shall be performed in any order which is satisfactory to the Government.

Table II - Group B Inspection

Inspection	Req. Para.	Insp. Para.	AQL
Input Power	3.13.1	4.8.1	6.5% for combined group
Connector Resistance	3.13.2	4.8.2	
Circuit Breaker, Trip-Out	3.13.6	4.8.6	
Interchangeability	3.14	4.9	6.5%

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4.5.4 Group C inspection. - This inspection shall be as listed in Table III and shall be performed on sample unit(s) that have been subjected to and met Group A and Group B inspections. No defects shall be allowed. Sample unit(s) shall be selected in accordance with 4.5.4.1.

Table III - Group C Inspection

Inspection	Rqt Para	Insp Para
Subgroup I:		
Temperature	3.12.1	4.7.1
Altitude	3.12.2	4.7.2
Relative Humidity	3.12.3	4.7.3
Vibration	3.12.5	4.7.5
Subgroup II:		
Sand and Dust	3.12.4	4.7.4
Benchhandling	3.12.6.1	4.7.6.1
Equipment	3.12.6.2	4.7.6.2
Subgroup III:		
Fungus	3.12.7	4.7.7
Salt Fog	3.12.8	4.7.8
Explosive Conditions	3.12.9	4.7.9

4.5.4.1 Sampling for inspection. - Units shall be selected at random for each group of inspections in Table III as specified in 4.5.4.1.1, through 4.5.4.1.3. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot. Selection of these samples shall be as specified and within 30 calendar days after selection these samples shall have been subjected to and passed the Group C inspections.

4.5.4.1.1 Subgroup I. - For this subgroup, two units from the first twenty-five (25) production units shall be selected for the first Group C inspection. For subsequent Group C inspection, two units from each successive 250, or two per month if less than 250 or fractions thereof are produced monthly shall be selected.

4.5.4.1.2 Subgroup II. - For this subgroup, two units (these units may be the units selected for above subgroup 1), from the first twenty-five (25) production units shall be selected for the first Group C inspection. For subsequent Group C inspection, two units from each 400 or fraction thereof or two every three months if less than 400 or fraction thereof are produced during three successive months, shall be selected.

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4.5.4.1.3 Subgroup III. - For this subgroup, two units from the first twenty-five (25) production units shall be selected for the first Group C inspection. For subsequent Group C inspection, two units shall be selected when one-half of the units on order have been accepted by the Government QA representative.

4.5.5 No-compliance. - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, and applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.6 Reinspection of conforming group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order, provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass Group A inspection.

4.6 Burn-in, debugging. - Each unit on order shall be subjected to and meet the burn-in test in accordance with Drawing SC-A-678003. Prior to production the contractor shall submit to the procuring activity a complete description of his debugging facility, and details of test procedures for approval.

4.7 Service conditions test. - The units shall be tested in accordance with Specification MIL-T-5422 to prove compliance with the environmental service conditions in Specification MIL-E-5400 for Class 2 equipment as specified. When a service condition test of MIL-T-5422 requires a performance check of the units, then the tests specified in paragraphs 4.8.6 and 4.10 shall be performed. The container, cover and hardware shall also be checked for corrosion after the conclusion of the temperature, humidity, fungus and salt spray tests.

4.7.1 Temperature. -

4.7.1.1 Operating. - Unit(s) shall meet the operating test of MIL-T-5422 for compliance with 3.12.1.1.

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4.7.1.2 Non-operating. - Unit(s) shall meet the non-operating tests of MIL-T-5422 for compliance with 3.12.1.2.

4.7.2 Altitude. - Unit(s) shall meet the operating and non-operating tests of MIL-T-5422 for compliance with 3.12.2.

4.7.3 Relative humidity. - The unit(s) shall be subjected to the humidity test of Specification MIL-T-5422 to determine compliance with 3.12.3.

4.7.4 Sand and dust. - The unit(s) shall be subjected to the sand and dust test of Specification MIL-T-5422 for compliance with 3.12.4.

4.7.5 Vibration. - The unit(s) shall be subjected to the vibration test of MIL-T-5422 for compliance with 3.12.5.

4.7.6 Shock. -

4.7.6.1 Benchhandling. - The unit(s) shall be assemblied for servicing and placed in a suitable position for servicing on a solid two-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

a. Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axis for a total of four drops.

b. Repeat a, with the assembly resting on other faces, until it has been dropped a total of four times on each face on which the assembly could be practically placed for servicing. Tests shall be performed before and after shock to insure compliance with paragraph 4.10 at supply voltage of 27.5 volts.

4.7.6.2 Equipment. - The unit(s) shall be subjected to the shock tests of Specification MIL-T-5422. These tests shall include the 30 G Clash Safety Test.

4.7.7 Fungus. - The unit(s) shall be subjected to 28 days of the fungus test specified in Method 508 of MIL-STD-810. For gasket or other non-hermetically sealed equipment, the parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall also be sprayed. The unit(s) shall be subjected to the test in this open condition and as specified for compliance with 3.12.7.

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4.7.8 Salt fog. - The unit(s) shall be exposed in its normal operating condition to salt fog in accordance with Method 509 of MIL-STD-810 for 48 hours. The unit(s) shall not be opened and shall not be required to be operated. The unit(s) shall meet the requirement of 3.12.8.

4.7.9 Explosive conditions. - Unit(s) shall be tested for explosion proof in accordance with MIL-T-5422 for compliance with 3.12.9.

4.8 Electrical tests. -

4.8.1 Input power. - The input power requirements of the unit(s) shall be tested for compliance with 3.13.1.

4.8.2 Connector resistance. - Utilizing the ohm meter section of a VTVM and with the unit de-energized, resistance between the indicated points of the connector pins shall be measured for compliance with 3.13.2.

4.8.3 Lamp operation. - Connect two test lamps MS-25237-327, or equal, with one lamp across Pin D and C and the other K and J. Jumper Pin B and G and connect a resistor (14 to 14.5 ohms) across Pins F and B. Apply 27.5 volts dc to Pins A, H and E. The lamps shall light as specified in 3.13.3.

4.8.4 Circuit breaker, voltage drop. - With a resistor (14 to 14.5 ohm) connected across Pins F and B, apply 27.5 volts dc to Pins A and E of the connector and measure the voltage drop across the "Power-On" switch for compliance with 3.13.4.

4.8.5 Switch, voltage drop. - Apply 27.5 volts dc to Pins A and E of the connector and measure the voltage from Pin E to Pin C, and from Pin E to Pin J, Pin E to Pin B for compliance with 3.13.5.

4.8.6 Circuit breaker, trip-out. - With the "Power-On" switch in "On" position apply sufficient dc voltage to Pin A and E of connector (J1). Apply the required load for the time required for the circuit breaker to "trip" to overload. The switch shall be reset to "On" position for compliance with 3.13.6.

4.9 Inspection for dimensional interchangeability. - The dimensions listed below shall be gaged or measured to determine compliance to the physical interchangeability requirements of 3.14. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases and covers, when such dimensions affect mating of parts.

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b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of fasteners or separable parts which must mate, such as cases, covers, and mounting devices.

d. Location of connectors, locking pins, fasteners, slides and mountings which receive parts.

4.10 Operational inspection. - The unit(s) shall be operated with 27.5 volts input as an operational unit. Switches and lights shall be operated to determine that they perform each required function. The requirements of 3.16 shall be met.

4.11 Visual and mechanical inspection. - Unit(s) shall be examined for the defects listed in MIL-STD-252 for compliance with 3.18.

4.12 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the packaging requirements.

4.13 Rough handling test (preparation for delivery). - When rough handling test in accordance with Specification MIL-P-116 is required by the contract the operational inspection test 4.10 shall be conducted to determine freedom from operational malfunction caused by rough handling.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be Level A or C as specified. (See Section 6.2).

5.1.1 Level A. -

5.1.1.1 Cleaning. - Each unit shall be cleaned in accordance with Process C-1 of MIL-P-116.

5.1.1.2 Drying. - Each unit shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116, as specified herein.

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5.1.1.4.1 Technical literature. - Each technical literature shall be packaged Method IC-1.

5.1.1.4.2 Control Indicator C-8157( )/ARC. - Units shall be packaged Method III, as follows; cushion each control by wrapping in paperboard cushioning material Type I, conforming to PPP-P-291. Seal the cushioning material with gummed paper tape conforming to PPP-T-45. Place the cushioned control within a folding paperboard box conforming to PPP-B-566 or set up paperboard box conforming to PPP-B-676. Box closure shall be in accordance with the appendix of the applicable box specification.

5.1.1.4.3 Intermediate container. - A quantity of controls packaged as specified in 5.1.1.4.2, bearing the same stock number shall be placed within a fiberboard box conforming to PPP-B-636, W5c or W6c, not to exceed 20 pounds. Box closure shall be in accordance with the appendix of the box specification.

5.1.2 Level C. - The units shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. - Packing shall be Level A, B or C as specified. (See 6.2).

5.2.1 Level A. -

5.2.1.1 Palletized load. - A quantity of units packaged as specified in 5.1.1.4.3 shall be placed on a pallet, Load Type I, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accomadate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather resistant W5s, the load shall be banded to the pallet by strapping.

5.2.1.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of the units packaged as specified in 5.1.1.4.3 shall by packed within a close-fitting fiberboard box conforming to PPP-B-640, Class 2, Grade A, Style E or MIL-B-4329l. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be made by nonmetallic strapping or pressure sensitive tape banding conforming to PPP-S-760, Type II or PPP-T-97, Type IV respectfully. Selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.2 Level B. - Units packaged as specified in 5.1, shall be packed as specified in 5.2.1.

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5.2.3 Level C. - Units packaged as specified in 5.1, shall be packed as specified in 5.2.1, except that the pallet, fiberboard boxes and fiberboard caps shall be of the type size and kind commonly used for the purpose and shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - Each unit is intended for use by the pilot, copilot or radio operator to provide control of security equipment when operated with various communication equipments installed in either fixed or rotor wing aircraft.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Levels of preservation, packaging, and packing required. (Level A, Level B, or Level C).

c. The specific paragraphs of Section 5 which are applicable to the particular procurement.

d. Preproduction inspection:

(1) Four (4) sample units of each item cited in Section 1 are generally required so that lengthly environmental tests can be completed on specified sample units while complete performance measurements can be made on one sample unit. (See 3.3)

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6.2 Ordering data (Cont'd)

(2) Preproduction pack(s) as follows:
Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon (including rough handling).

e. Marking and shipping of sample.

f. Place of final inspection.

g. Technical literature required.

h. Quantity of spare parts required.

i. Submission of statement of treatment referenced in 3.11.4 should be sixty (60) days after award of contract. This statement should be submitted to the contracting officer.

6.3 Location of operational inspection. - It is desirable that the operational inspection 4.10 be performed at a location that will minimize handling (which might cause damage to the units) after this inspection. It is recommended that the entire lot (including the sample units) be inspected immediately prior to packaging.

6.4 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: C-8157W/ARC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.5 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
Army EL

Preparing activity:
Army EL
Project: 5821 -A070

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-55688(EL) CONTROL INDICATOR C-8157( )/ARC

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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